
KiranaChain: A Multi-Agent Negotiation Benchmark with Informal Credit Dynamics and Code-Switched Dialogue for Emerging Market AI Research

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Abstract

The rapid advancement of artificial intelligence has been fueled by the availability of large-scale benchmark datasets that capture complex decision-making environments. However, existing public benchmarks remain predominantly centered on centralized digital ecosystems and often fail to represent the decentralized, relationship-driven economic structures that characterize emerging markets. In particular, the absence of datasets modeling informal credit systems, supply-chain uncertainty, multilingual negotiation behavior, and repeated economic interactions limits the development and evaluation of next-generation autonomous agents operating in real-world commercial environments.

This paper introduces **KiranaChain**, a large-scale multi-agent negotiation benchmark designed to model decentralized retail commerce within the context of emerging-market economies. The benchmark comprises **1,000,000 negotiation-turn observations** organized into **125,000 complete multi-agent trajectories**, each representing an eight-stage bargaining process between wholesale distributors and independent Kirana retailers. KiranaChain integrates a **40-dimensional state space** encompassing agent attributes, geographic context, informal credit exposure, trust dynamics, liquidity constraints, environmental disruptions, supply-chain intelligence, negotiation strategies, and multilingual code-switched dialogue telemetry.

To achieve realistic behavioral diversity, the dataset is generated through a scientifically grounded simulation framework that combines Gaussian copula dependency modeling, Cholesky-based correlation preservation, non-stationary Markov negotiation processes, and Gaussian Process stochastic perturbation mechanisms. The resulting benchmark supports a broad spectrum of research domains, including offline reinforcement learning, multi-agent systems, graph representation learning, agentic AI, conversational intelligence, credit-risk estimation, supply-chain optimization, behavioral economics, and multilingual language modeling.

Comprehensive validation procedures demonstrate structural integrity, statistical consistency, trajectory-level independence, and distributional robustness across training, validation, and test partitions. Furthermore, the benchmark introduces novel opportunities for studying trust-aware autonomous negotiation, informal credit allocation mechanisms, decentralized economic coordination, and multilingual commercial reasoning under uncertainty.

By bridging the gap between artificial intelligence research and decentralized retail economies, KiranaChain establishes a new benchmark paradigm for developing economically grounded, socially aware, and strategically adaptive AI systems. We anticipate that the dataset will serve as foundational infrastructure for future research in autonomous negotiation, economic intelligence, and emerging-market AI ecosystems.

Keywords: Multi-Agent Systems, Negotiation Intelligence, Offline Reinforcement Learning, Agentic Artificial Intelligence, Large Language Models, Graph Neural Networks, Economic Intelligence, Informal Credit Systems, Supply Chain Analytics, Behavioral Economics, Code-Switched Dialogue, Multilingual NLP, Decentralized Commerce.

Dataset Availability

Hugging Face: <https://huggingface.co/datasets/Atlas-AI-Labs/KiranaChain>

DOI: <https://doi.org/10.57967/hf/9015>

GitHub Repository: <https://github.com/dspraneeth07/KiranaChain>

1 Introduction

1.1 The Decentralized Retail Landscape

Recent advances in machine learning, reinforcement learning, graph learning, and large language models have been driven by benchmark-centric research ecosystems [1, 2, 5, 6]. However, most existing datasets focus on centralized digital environments, including recommendation systems, online platforms, and simulated agent interactions, limiting their applicability to decentralized economic settings.

The Indian Kirana ecosystem represents one of the world’s largest decentralized retail networks, consisting of independent retailers, distributors, and informal supply-chain relationships. Commercial decisions are shaped by bargaining, credit dependencies, inventory constraints, and localized market conditions. Unlike structured retail environments, outcomes emerge through repeated interactions among heterogeneous agents operating under uncertainty.

From an AI perspective, such ecosystems naturally align with multi-agent systems, reinforcement learning, and graph-based modeling frameworks [9, 10, 11, 12]. In addition, commercial communication frequently involves multilingual and code-switched dialogue, creating challenges for conversational intelligence and language modeling. Although recent foundation models have demonstrated strong reasoning and decision-making capabilities [7, 8, 20], the lack of benchmarks grounded in decentralized economic interactions remains a significant limitation. Consequently, there is a need for datasets that jointly capture negotiation dynamics, informal credit behavior, supply-chain uncertainty, and multilingual communication.

1.2 The Foundational Data Crisis

Large-scale datasets have played a central role in advancing artificial intelligence across language processing, graph learning, and reinforcement learning [1, 2, 3, 4]. Similarly, negotiation and dialogue benchmarks such as DealOrNoDeal, CraigslistBargain, and MultiWOZ have enabled progress in conversational agents and strategic interaction modeling [16, 17, 18, 19]. Despite these contributions, existing benchmarks remain insufficient for modeling decentralized commercial ecosystems.

First, most datasets rely on independently and identically distributed observations and fail to capture long-horizon negotiation processes and evolving agent relationships. Second, informal credit allocation, trust accumulation, repayment behavior, and financial risk are rarely represented despite their importance in emerging-market commerce. Third, supply-chain disruptions, logistics uncertainty, and environmental shocks are generally absent from benchmark design. Fourth, multilingual and code-switched commercial interactions remain underrepresented despite their prevalence in real-world retail environments [5, 7, 8]. Finally, reinforcement learning, graph learning, dialogue modeling, and credit-risk analysis are typically studied using separate datasets, limiting interdisciplinary research.

As a result, current benchmark ecosystems inadequately capture:

- Sequential bargaining and negotiation trajectories.
- Informal credit relationships and trust dynamics.
- Supply-chain disruptions and environmental uncertainty.
- Multilingual and code-switched commercial dialogue.
- Evolving economic relationships among agents.

Addressing these limitations requires benchmark infrastructures specifically designed for decentralized economic intelligence and trajectory-based decision-making.

1.3 Our Contributions

To address these challenges, we introduce **KiranaChain**, a large-scale benchmark for decentralized retail negotiations, informal credit systems, supply-chain interactions, and multilingual commercial communication. The benchmark integrates trajectory-based decision processes with structured economic and conversational telemetry, enabling research across multiple AI domains.

The main contributions of this work are summarized as follows:

1. **Novel Benchmark Design:** Introduction of KiranaChain, a benchmark tailored for decentralized retail ecosystems and economic interaction modeling.
2. **Large-Scale Trajectory Dataset:** Provision of approximately **1,000,000 negotiation-turn observations** organized into **125,000 complete trajectories** for sequential learning and policy evaluation.
3. **Informal Credit Modeling:** Inclusion of trust formation, repayment behavior, liquidity constraints, and credit-risk indicators within negotiation trajectories.
4. **Multilingual Dialogue Telemetry:** Integration of code-switched commercial conversations to support conversational AI and language-model research.
5. **Multi-Paradigm Benchmarking:** Support for reinforcement learning, multi-agent systems, graph neural networks, credit-risk modeling, behavioral economics, and large language models [2, 3, 4, 5].
6. **Reproducible Simulation Framework:** Generation of trajectories through a structured simulation pipeline incorporating correlated economic variables, negotiation dynamics, and stochastic environmental factors.

Collectively, KiranaChain provides a unified benchmark for studying decentralized economic intelligence, autonomous negotiation, and decision-making under uncertainty, while enabling reproducible evaluation across multiple AI research communities.

2 Related Work and Ontological Scope

The development of benchmark datasets has historically played a central role in accelerating scientific progress across artificial intelligence, machine learning, natural language processing, and decision intelligence. Foundational advances in deep reinforcement learning [10, 9, 11, 12], graph representation learning [2, 3, 4], large language models [1, 5, 6, 7], and autonomous reasoning systems [20] have been enabled by benchmark-driven evaluation paradigms.

However, despite remarkable progress, existing benchmark ecosystems remain fragmented. Negotiation datasets focus primarily on dialogue generation, reinforcement learning environments emphasize policy optimization, graph datasets target relational learning, and credit-risk datasets concentrate on financial prediction tasks. Consequently, no unified benchmark currently captures the interaction between negotiation behavior, informal credit allocation, multilingual communication, supply-chain uncertainty, and dynamic economic relationships. This section situates KiranaChain within the broader landscape of negotiation intelligence, financial analytics, and multilingual conversational systems.

2.1 Multi-Agent Negotiation Benchmarks

Negotiation constitutes one of the most fundamental forms of strategic interaction among autonomous agents. Early research in automated negotiation primarily focused on utility optimization and game-theoretic bargaining mechanisms. More recent advances have leveraged deep learning and language models to study negotiation as a sequential decision-making problem involving both strategic planning and natural language communication.

Among the most influential benchmarks, *DealOrNoDeal* introduced an end-to-end framework for learning negotiation strategies directly from dialogue interactions [16]. The dataset demonstrated that neural agents could learn bargaining policies while simultaneously generating coherent negotiation language. Building upon this paradigm, *CraigslistBargain* expanded the scope of negotiation research by introducing richer conversational exchanges and more realistic buyer-seller interactions [18]. Similarly, subsequent work explored the decoupling of strategic planning from language generation, highlighting the importance of explicit reasoning mechanisms within negotiation agents [17].

Parallel developments in task-oriented dialogue systems produced large-scale datasets such as *MultiWOZ*, which significantly advanced conversational intelligence research [19]. Although such datasets provide extensive multi-turn interactions, they primarily focus on information retrieval and task completion rather than economic decision-making, bargaining, or trust formation.

Despite their contributions, existing negotiation benchmarks exhibit several limitations when viewed through the lens of decentralized commerce. First, most datasets model isolated transactions rather than persistent economic relationships. Second, trust dynamics, informal credit allocation, and long-term financial dependencies remain absent. Third, existing benchmarks rarely incorporate external environmental factors such as logistics disruptions or supply-chain uncertainty. Finally, multilingual and code-switched communication patterns are largely underrepresented.

Consequently, while prior datasets have substantially advanced negotiation research, they do not fully capture the complexity of decentralized retail ecosystems characterized by repeated interactions, credit dependencies, and evolving economic relationships.

2.2 Econometric and Credit Risk Datasets

Financial risk assessment has traditionally been studied through structured econometric datasets containing demographic attributes, repayment histories, transaction records, and credit-scoring indicators. Benchmark datasets such as FICO challenge datasets, LendingClub repositories, and various banking risk corpora have facilitated significant advances in credit-risk prediction, default estimation, and explainable financial decision-making.

These datasets have enabled the application of machine learning techniques ranging from classical statistical models to modern deep learning architectures for predicting loan performance and borrower reliability. However, most existing financial datasets are derived from formal institutional lending environments governed by standardized regulations and well-defined contractual structures.

In contrast, informal credit ecosystems commonly observed in emerging markets operate through trust-based relationships rather than formal financial instruments. Credit decisions often depend upon social reputation, historical interactions, geographic proximity, and long-term commercial dependencies. Such characteristics introduce dynamic behavioral components that differ substantially from conventional banking datasets.

Furthermore, existing credit-risk benchmarks typically model financial outcomes as static prediction problems. They rarely capture negotiation processes that precede credit allocation decisions, nor do they represent how trust evolves through repeated interactions. As a result, the interaction between economic bargaining, credit extension, and repayment behavior remains largely unexplored within current benchmark infrastructures.

KiranaChain addresses this gap by integrating informal credit dynamics directly into negotiation trajectories, enabling researchers to study credit-risk estimation, trust-aware decision-making, and economic relationship modeling within a unified benchmark environment.

2.3 Code-Switched Linguistic Telemetry

Recent advances in natural language processing have been driven by transformer-based architectures and large-scale pretraining paradigms [1, 5, 6]. Foundation models such as GPT-series and LLaMA-family systems have demonstrated remarkable capabilities across reasoning, generation, and instruction-following tasks [7, 8]. More recently, reasoning-oriented frameworks such as ReAct have combined language understanding with decision-making and action planning [20].

Despite these advances, multilingual and code-switched communication remains an open challenge for language technologies. In many emerging-market environments, commercial conversations naturally alternate between multiple languages, dialects, and domain-specific terminology. Such linguistic behavior introduces semantic ambiguities, contextual dependencies, and pragmatic reasoning requirements that are not adequately represented in predominantly monolingual datasets.

Existing dialogue benchmarks largely focus on English-language interactions and standardized conversational structures. Consequently, they provide limited support for studying multilingual negotiation behavior, commercial discourse strategies, and language-grounded economic reasoning.

From a benchmark design perspective, code-switched dialogue serves not only as a linguistic phenomenon but also as an important source of behavioral information. Negotiation intent, trust signals, urgency indicators, and risk perceptions are frequently expressed through language choices that vary across cultural

and regional contexts. Capturing such phenomena is therefore essential for developing economically aware conversational agents capable of operating in real-world decentralized markets.

KiranaChain incorporates multilingual and code-switched dialogue telemetry as a first-class component of the benchmark, enabling future research on language-grounded negotiation intelligence, multilingual agent systems, and economically contextualized large language models.

2.4 Comparative Benchmark Positioning

To better contextualize the contribution of KiranaChain, Table 1 compares representative benchmark datasets across four core dimensions: negotiation capability, credit modeling, trajectory structure, and multilingual code-switched communication.

Dataset	Negotiation	Credit	Trajectory	Code-Switch
DealOrNoDeal [16]	✓	–	Partial	–
CraigslistBargain [18]	✓	–	Partial	–
MultiWOZ [19]	–	–	✓	–
KiranaChain	✓	✓	✓	✓

Table 1: Comparative positioning of KiranaChain relative to representative negotiation and dialogue benchmarks.

The comparison highlights a significant gap within existing benchmark ecosystems. While prior datasets provide valuable resources for dialogue generation and negotiation research, none simultaneously model negotiation behavior, informal credit systems, trajectory-based economic interactions, and multilingual code-switched communication. KiranaChain is designed specifically to address this intersection, thereby enabling a broader class of research problems spanning reinforcement learning, agentic AI, graph learning, economic intelligence, and conversational decision-making.

3 Telemetry Acquisition and Generation Framework

A defining challenge in constructing economically grounded benchmark datasets is the preservation of realistic inter-variable dependencies while simultaneously maintaining scalability, reproducibility, and statistical controllability. Unlike conventional datasets derived from isolated transactional records, KiranaChain models decentralized retail ecosystems as dynamic negotiation environments characterized by heterogeneous agent behaviors, informal credit relationships, supply-chain externalities, and multilingual conversational interactions.

To achieve this objective, the benchmark is generated through a structured telemetry acquisition framework consisting of four interconnected components: (i) high-dimensional dependency modeling, (ii) non-stationary negotiation state transitions, (iii) temporally autocorrelated behavioral perturbations, and (iv) trajectory-level validation and quality assurance. The resulting framework enables the synthesis of economically plausible negotiation trajectories while preserving mathematical transparency and reproducibility. [9, 10, 11]

3.1 KiranaChain: High-Dimensional Dependency Modeling

Real-world commercial ecosystems exhibit strong correlations among financial, environmental, and behavioral variables. For example, liquidity stress influences default risk, trust affects credit allocation, and supply-chain disruptions impact negotiated pricing. Modeling such dependencies using independent sampling procedures produces unrealistic state configurations and weakens downstream learning tasks.

To preserve realistic dependency structures, KiranaChain employs a Gaussian Copula framework. Gaussian copulas have become a standard mechanism for generating correlated multivariate systems while allowing heterogeneous marginal distributions. The approach enables explicit control over target Pearson correlations while maintaining flexibility in variable-specific distributions.

Let

$$Z \sim \mathcal{N}(0, \Sigma)$$

denote a multivariate Gaussian random vector generated from a positive semi-definite correlation matrix Σ . A Cholesky decomposition

$$\Sigma = LL^T$$

is first applied to construct correlated latent variables. These latent variables are subsequently transformed through the standard normal cumulative distribution function and variable-specific inverse cumulative distributions:

$$X = F^{-1}(\Phi(Z)) \quad (1)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function and $F^{-1}(\cdot)$ represents the inverse marginal distribution associated with a target variable.

Within KiranaChain, this framework is used to generate correlated financial attributes including trust scores, credit limits, liquidity stress indices, repayment velocities, default probabilities, and informal interest rates. Similarly, environmental variables such as monsoon disruption indices, perishability decay rates, fuel-price volatility, and inflation indicators are generated using an independent environmental copula block. The resulting dependency structure produces economically coherent state configurations while preserving statistical diversity across the benchmark. [3, 2, 4]

Furthermore, copula-based generation permits the explicit encoding of domain knowledge. For example, trust scores exhibit strong negative correlation with default probabilities, while liquidity stress demonstrates positive correlation with repayment delays and credit risk. Such relationships are difficult to reproduce through naive sampling procedures yet emerge naturally within the copula framework.

3.2 Non-Stationary State Transitions

Negotiation processes are inherently sequential and evolve through repeated strategic interactions. Conventional Markovian environments often assume stationary transition dynamics, implying that transition probabilities remain constant throughout an episode. However, real-world bargaining behavior rarely satisfies this assumption.

As negotiations progress, settlement likelihood typically increases, strategic flexibility changes, and external pressures accumulate. To capture these dynamics, KiranaChain employs a non-stationary turn-conditioned Markov negotiation engine.

The negotiation state space is defined as

$$\mathcal{S} = \{\text{In-Progress, Settled, Walked-Away, Credit-Denied}\}$$

where terminal states are absorbing.

State evolution is governed by

$$P(s_{t+1} \mid s_t, t) \quad (2)$$

which explicitly conditions transition probabilities on both the current state and negotiation turn index. Unlike stationary Markov chains, the transition operator evolves throughout the bargaining process, reflecting increasing settlement pressure and changing behavioral incentives.

Several economic variables further modulate transition behavior, including:

- Historical trust score,
- Liquidity stress index,
- Outstanding informal credit,
- Repayment velocity,

- Supply-chain disruption intensity,
- Price convergence dynamics.

The resulting negotiation trajectories exhibit realistic bargaining patterns including concession progression, settlement convergence, walk-away events, and credit-denial scenarios. Such dynamics create a rich sequential environment suitable for offline reinforcement learning, policy optimization, multi-agent coordination, and strategic reasoning research. [10, 11, 12, 16, 17]

From a benchmark-design perspective, the use of non-stationary transitions substantially increases behavioral diversity while preserving reproducibility. This enables the generation of large-scale trajectory datasets without sacrificing temporal realism.

3.3 Temporally Autocorrelated Perturbations

Human negotiation behavior is rarely deterministic. Decisions are influenced by unobserved factors including emotion, urgency, bounded rationality, information asymmetry, and local market conditions. Consequently, realistic benchmark generation requires mechanisms capable of introducing controlled stochasticity while preserving temporal consistency.

To model such effects, KiranaChain incorporates Gaussian Process (GP) perturbation mechanisms into every negotiation trajectory. Unlike independently sampled noise, Gaussian Processes generate temporally correlated perturbations that more closely resemble human behavioral variation.

The covariance structure is defined through a squared-exponential kernel:

$$k(t_i, t_j) = \exp\left(-\frac{(t_i - t_j)^2}{2l^2}\right) \quad (3)$$

where t_i and t_j denote negotiation turns and l represents the characteristic length scale controlling temporal smoothness.

The kernel matrix

$$K_{ij} = k(t_i, t_j)$$

defines a multivariate Gaussian distribution over perturbation vectors:

$$\mathbf{g} \sim \mathcal{N}(0, K)$$

from which trajectory-specific noise realizations are sampled.

These perturbations are subsequently injected into negotiated price paths, sentiment trajectories, and behavioral telemetry variables. Because neighboring turns exhibit strong covariance, perturbations evolve smoothly across an episode rather than producing unrealistic independent fluctuations.

The incorporation of Gaussian Process noise serves several objectives:

1. Modeling bounded human rationality.
2. Capturing latent market frictions.
3. Simulating measurement uncertainty.
4. Increasing behavioral diversity.
5. Preventing deterministic trajectory collapse.

Such mechanisms are particularly important for reinforcement learning benchmarks, where overly deterministic environments can encourage brittle policies that fail to generalize beyond narrow state distributions. [13, 14, 15]

3.4 Integrated Telemetry Pipeline

The complete KiranaChain telemetry framework integrates dependency modeling, sequential state transitions, and stochastic perturbation mechanisms into a unified generation architecture. The pipeline operates through the following stages:

1. Agent instantiation and geographic assignment.
2. Financial telemetry generation via Gaussian copulas.
3. Environmental telemetry generation via Gaussian copulas.
4. Price initialization and bargaining setup.
5. Non-stationary Markov negotiation simulation.
6. Concession trajectory evolution.
7. Gaussian Process perturbation injection.
8. Behavioral posture assignment.
9. Multilingual dialogue synthesis.
10. Sentiment scoring and annotation.
11. Parquet streaming and storage.
12. Validation and quality assurance.

Collectively, these components produce approximately one million negotiation-turn observations organized into 125,000 complete trajectories. The resulting benchmark combines economic realism, statistical rigor, and computational reproducibility, thereby providing a scalable research platform for reinforcement learning, multi-agent systems, graph intelligence, conversational AI, and decentralized economic modeling.

4 Dataset Scale, Taxonomy and Topology

Large-scale benchmark datasets have historically served as foundational catalysts for progress in machine learning, graph intelligence, reinforcement learning, and conversational AI. Datasets such as ImageNet transformed visual learning, while graph benchmarks and large-scale language corpora accelerated advances in representation learning, reasoning, and autonomous decision systems [2, 3, 4, 5, 7, 8]. Despite these developments, existing public benchmarks predominantly focus on centralized digital ecosystems and fail to represent decentralized economic interactions that characterize large segments of emerging-market commerce.

KiranaChain addresses this gap through the introduction of a trajectory-centric benchmark consisting of 1,000,000 negotiation-turn observations organized into 125,000 complete distributor–retailer negotiation trajectories. Each trajectory contains eight temporally ordered bargaining stages and is represented through a structured 40-dimensional state space integrating economic, behavioral, environmental, logistical, and conversational variables. The resulting benchmark establishes one of the first large-scale research environments designed specifically for decentralized retail intelligence, informal credit ecosystems, supply-chain dynamics, and multilingual negotiation modeling.

4.1 Comprehensive Mathematical Taxonomy

A central design principle of KiranaChain is the decomposition of complex economic interactions into interpretable and semantically coherent feature subsystems. Rather than treating observations as independent tabular records, the benchmark organizes variables according to their functional role within the negotiation process.

The complete state representation is defined as

$$\mathbf{x}_t = [\mathbf{x}^A, \mathbf{x}^F, \mathbf{x}^S, \mathbf{x}^N]$$

where

- $\mathbf{x}^A$ denotes Agent Intelligence variables,
- $\mathbf{x}^F$ denotes Financial Intelligence variables,
- $\mathbf{x}^S$ denotes Supply-Chain Intelligence variables,
- $\mathbf{x}^N$ denotes Negotiation Intelligence variables.

Collectively, these components define a structured economic state representation suitable for reinforcement learning, graph representation learning, sequence modeling, and multi-agent reasoning tasks.

Table 2: Hierarchical Taxonomy of KiranaChain Feature Architecture

Block	Features	Columns	Primary Function
Agent Intelligence	8	1–8	Identity, geography, commodity context, market location
Financial Intelligence	10	9–18	Trust modeling, informal credit allocation, liquidity stress, risk estimation
Supply Chain Intelligence	10	19–28	Environmental uncertainty, logistics disruptions, inflation, perishability
Negotiation Intelligence	12	29–40	Price evolution, bargaining behavior, dialogue telemetry, outcomes

This taxonomy establishes a modular representation that enables both subsystem-specific and end-to-end learning paradigms. Financial variables may support credit-risk estimation, supply-chain variables enable disruption forecasting, while negotiation variables facilitate sequential decision-making and dialogue policy learning.

From a representation-learning perspective, the taxonomy naturally supports heterogeneous feature fusion architectures, graph neural networks, transformers, and multi-modal agent systems [1, 2, 4].

4.2 Dataset Scale and Structural Characteristics

The benchmark contains

$$N = 1,000,000$$

individual observations grouped into

$$M = 125,000$$

complete negotiation trajectories.

Each trajectory contains a fixed temporal horizon

$$T = 8$$

turns, producing a trajectory-structured environment rather than a collection of independent samples. Consequently, KiranaChain may be interpreted simultaneously as:

1. A supervised learning benchmark.
2. A sequential decision-making benchmark.

3. A reinforcement learning environment.
4. A multi-agent interaction dataset.
5. A graph-construction benchmark.
6. A multilingual conversational corpus.

This multi-paradigm design substantially expands the range of scientific questions that can be investigated using a single benchmark resource.

4.3 Structural Splitting Mechanics

Dataset partitioning is performed at the trajectory level rather than the observation level. This design choice is essential for preserving causal consistency and preventing temporal information leakage.

Formally,

$$\mathcal{D} = \mathcal{D}_{train} \cup \mathcal{D}_{val} \cup \mathcal{D}_{test} \quad (4)$$

subject to

$$\mathcal{D}_{train} \cap \mathcal{D}_{val} = \emptyset$$

$$\mathcal{D}_{train} \cap \mathcal{D}_{test} = \emptyset$$

$$\mathcal{D}_{val} \cap \mathcal{D}_{test} = \emptyset$$

at the trajectory level.

The benchmark adopts an 80/10/10 partitioning strategy:

Table 3: Trajectory-Level Benchmark Partitions

Split	Rows	Trajectories	Share
Train	800,000	100,000	80%
Validation	100,000	12,500	10%
Test	100,000	12,500	10%

Unlike conventional row-level splitting, this approach guarantees that no negotiation episode appears across multiple partitions. Consequently, models must generalize to entirely unseen bargaining trajectories rather than exploiting memorized temporal patterns.

This property is particularly important for offline reinforcement learning, policy optimization, trajectory forecasting, and sequence modeling systems [9, 10, 11, 12].

4.4 Macroeconomic Network Graphs

Beyond trajectory modeling, KiranaChain can be interpreted as a dynamic economic graph.

The evolving interaction structure is represented by

$$G_t = (V, E_t, W_t) \quad (5)$$

where:

- V denotes distributor and Kirana retailer agents,
- E_t denotes active transactional relationships at time t ,

- W_t denotes weighted edge attributes.

Edge weights encode multiple economic dimensions including

- Trust relationships,
- Credit exposure,
- Outstanding liabilities,
- Transaction frequency,
- Settlement behavior,
- Historical bargaining outcomes.

Unlike static graph benchmarks, KiranaChain produces a temporally evolving relational structure in which both node states and edge characteristics continuously change across negotiation episodes.

The resulting graph environment enables investigation of several advanced graph-learning problems:

1. Link prediction,
2. Dynamic graph forecasting,
3. Trust propagation modeling,
4. Credit contagion analysis,
5. Community discovery,
6. Temporal graph representation learning,
7. Economic network resilience analysis.

These capabilities position KiranaChain at the intersection of graph intelligence and economic systems research, extending beyond traditional citation networks, social graphs, and recommendation datasets commonly studied within graph machine learning literature [2, 3, 4].

4.5 Multi-Agent Topological Interpretation

Each trajectory may additionally be formalized as a decentralized partially observable multi-agent interaction process.

At turn t , the environment state is represented by

$$s_t = \{x_t^A, x_t^F, x_t^S, x_t^N\}$$

while distributor and retailer agents execute actions

$$a_t^{(d)} \quad \text{and} \quad a_t^{(r)}$$

corresponding to pricing concessions, credit decisions, and negotiation strategies.

The resulting trajectory can therefore be represented as

$$\tau = (s_1, a_1, r_1, \dots, s_T, a_T, r_T)$$

which directly aligns with the Markov Decision Process formalism underlying modern reinforcement learning frameworks [9, 10].

This formulation enables KiranaChain to support:

- Offline Reinforcement Learning,

- Multi-Agent Reinforcement Learning,
- Strategic Negotiation Agents,
- Agentic Economic Systems,
- Autonomous Decision-Making Research,
- Human-AI Negotiation Modeling.

Consequently, the benchmark bridges traditionally disconnected research communities spanning reinforcement learning, graph intelligence, behavioral economics, supply-chain analytics, and conversational AI.

4.6 Topological Significance

From a scientific benchmarking perspective, KiranaChain occupies a unique position among contemporary datasets. Existing benchmarks such as MultiWOZ focus on task-oriented dialogue [19], CraigslistBargain targets negotiation conversations [18], and recent language-model benchmarks emphasize reasoning and instruction following [5, 6, 20]. However, none simultaneously integrate:

1. Multi-agent economic interactions,
2. Informal credit dynamics,
3. Supply-chain uncertainty,
4. Dynamic graph structures,
5. Sequential bargaining trajectories,
6. Multilingual code-switched dialogue,
7. Reinforcement-learning-compatible state transitions.

The resulting topology transforms KiranaChain from a conventional tabular dataset into a unified economic intelligence benchmark capable of supporting next-generation research in autonomous negotiation, graph reasoning, economic decision intelligence, and agent-based systems.

5 Baseline Mathematical Formulations and Tasks

A defining characteristic of KiranaChain is its ability to support multiple machine learning paradigms within a unified benchmark environment. Unlike conventional datasets that target a single learning objective, KiranaChain simultaneously enables research in reinforcement learning, graph representation learning, strategic negotiation systems, conversational intelligence, credit-risk modeling, supply-chain analytics, and multi-agent decision making.

The benchmark therefore serves as a common evaluation platform for studying decentralized economic intelligence under uncertainty.

5.1 Offline Reinforcement Learning

Modern reinforcement learning systems rely upon structured state transitions that describe how actions influence future environmental conditions [9, 10, 11, 12].

Each KiranaChain negotiation trajectory naturally induces a Markov Decision Process (MDP):

$$M = (S, A, P, R, \gamma) \tag{6}$$

where:

- S denotes the negotiation state space,

- A denotes the action space,
- P represents transition dynamics,
- R denotes reward functions,
- γ is the discount factor.

The complete state vector at turn t is defined as

$$s_t = [x_t^A, x_t^F, x_t^S, x_t^N]$$

where agent, financial, environmental, and negotiation variables jointly characterize the economic environment.

Each observation can therefore be interpreted as a transition tuple

$$(s_t, a_t, r_t, s_{t+1}) \tag{7}$$

allowing direct application of offline reinforcement learning algorithms including Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), AlphaZero-style planning approaches, and model-based policy learning frameworks [10, 11, 12, 13, 14, 15].

Potential reward functions include:

- Settlement success,
- Transaction profitability,
- Credit-risk minimization,
- Inventory clearance,
- Long-term trust preservation.

This flexibility enables evaluation of both cooperative and competitive bargaining strategies under realistic economic constraints.

5.2 Multi-Agent Negotiation Learning

Each trajectory consists of interactions between two autonomous decision makers:

- Distributor Agent,
- Kirana Retail Agent.

The interaction process can be represented as

$$\tau = (s_1, a_1^{(d)}, a_1^{(r)}, r_1, \dots, s_T)$$

where

$$a_t^{(d)}$$

and

$$a_t^{(r)}$$

denote distributor and retailer actions respectively.

This structure closely aligns with contemporary multi-agent reinforcement learning environments and strategic negotiation systems [16, 17].

Research tasks supported by the benchmark include:

- Autonomous bargaining,
- Credit-aware negotiation,
- Cooperative settlement optimization,
- Strategic concession learning,
- Human-AI negotiation modeling.

Consequently, KiranaChain serves as a bridge between economic decision intelligence and multi-agent learning research.

5.3 Sequenced Structural Tokenization

Recent advances in large language models demonstrate that structured sequence representations significantly improve reasoning, planning, and instruction-following capabilities [5, 6, 7, 8, 20].

KiranaChain supports multiple sequence encoding paradigms.

5.3.1 Instruction-Tuning Representation

A trajectory may be transformed into an instruction-response pair:

State Context + Negotiation History $\rightarrow$ Recommended Action

enabling supervised fine-tuning of negotiation-oriented language models.

5.3.2 Dialogue-State Encoding

The conversational telemetry contained in `code_switched_dialogue_raw` and `llguidance_constrained_json_log` supports dialogue-state tracking and conversational reasoning tasks.

A structured dialogue representation may be expressed as

$$D = \{u_1, u_2, \dots, u_n\}$$

where u_i denotes individual negotiation utterances.

Such representations are analogous to dialogue-learning benchmarks including MultiWOZ and Craigslist-Bargain [19, 18].

5.3.3 Reward-Conditioned Trajectories

Following recent advances in trajectory-conditioned learning, negotiation histories may be serialized as

$$(s_1, a_1, r_1, \dots, s_T, a_T, r_T)$$

allowing language models to learn economic decision policies through sequence prediction.

This formulation provides a natural bridge between reinforcement learning and foundation model training.

5.4 Graph Representation Learning

Beyond sequential modeling, KiranaChain can be interpreted as a dynamic economic graph.

Let

$$G_t = (V, E_t, W_t) \tag{8}$$

represent the evolving transaction network.

Nodes correspond to distributors and retailers, while weighted edges encode:

- Trust relationships,

- Credit exposure,
- Transaction frequency,
- Repayment behavior,
- Historical settlement outcomes.

This representation naturally supports graph neural network architectures including Graph Convolutional Networks, GraphSAGE, and Graph Attention Networks [2, 3, 4].

Key benchmark tasks include:

- Link Prediction,
- Trust Propagation,
- Community Discovery,
- Credit Contagion Analysis,
- Dynamic Graph Forecasting,
- Node Classification,
- Edge Risk Estimation.

Table 4: Graph Learning Tasks Enabled by KiranaChain

Task	Objective	Potential Models
Link Prediction	Future trading relationships	GCN, GraphSAGE
Trust Propagation	Trust-score estimation	GAT, Temporal GNN
Community Discovery	Market clustering	Graph Embeddings
Credit Contagion	Default diffusion modeling	Temporal GNNs
Risk Forecasting	Edge-level credit risk	Graph Transformers

The graph formulation substantially broadens the scientific utility of the benchmark beyond traditional trajectory datasets.

5.5 Unified Decision Intelligence Formulation

A central design objective of KiranaChain is the unification of multiple learning paradigms within a single benchmark environment.

Let

$$\mathcal{B} = \{\tau_1, \tau_2, \dots, \tau_N\}$$

denote the complete benchmark consisting of

$$N = 125,000$$

negotiation trajectories.

Each trajectory is represented as

$$\tau_i = \{(s_1, a_1, r_1), (s_2, a_2, r_2), \dots, (s_T, a_T, r_T)\}$$

with

$$T = 8.$$

The benchmark therefore admits a generalized learning operator

$$\mathcal{F} : \mathcal{B} \rightarrow \mathcal{Y}$$

where

$$\mathcal{Y}$$

may correspond to:

- Actions (Reinforcement Learning),
- Policies (Offline RL),
- Dialogue Tokens (Language Modeling),
- Graph Embeddings (Graph Learning),
- Risk Scores (Credit Intelligence),
- Settlement Outcomes (Supervised Learning),
- Strategic Behaviors (Multi-Agent Systems).

Consequently, KiranaChain can be interpreted as a general-purpose decentralized economic intelligence benchmark rather than a narrowly specialized dataset.

5.6 Economic State Representation Learning

The benchmark provides a structured 40-dimensional state representation

$$s_t = [x_1, x_2, \dots, x_{40}]$$

partitioned into four semantically distinct information blocks:

$$s_t = \left[s_t^{(A)}, s_t^{(F)}, s_t^{(E)}, s_t^{(N)} \right]$$

where

- $s_t^{(A)}$ denotes agent intelligence,
- $s_t^{(F)}$ denotes financial intelligence,
- $s_t^{(E)}$ denotes environmental intelligence,
- $s_t^{(N)}$ denotes negotiation intelligence.

This decomposition supports representation-learning studies investigating how different information modalities contribute to economic decision quality.

Potential research questions include:

- Which information block contributes most to settlement prediction?
- Can latent economic embeddings improve negotiation policies?
- How does environmental uncertainty influence learned representations?
- Can trust dynamics emerge as latent variables?

Such questions remain largely unexplored in existing benchmark ecosystems.

5.7 Offline Policy Evaluation

One of the most challenging problems in reinforcement learning involves estimating policy quality without online interaction [9, 11, 12]. Given a target policy

$$\pi(a|s)$$

and historical trajectory set

$$\mathcal{D} = \{(s_t, a_t, r_t, s_{t+1})\},$$

the expected return is

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^T \gamma^t r_t \right].$$

KiranaChain supports:

- Importance Sampling,
- Weighted Importance Sampling,
- Doubly Robust Estimation,
- Fitted Q Evaluation,
- Conservative Offline RL.

The availability of complete trajectory information makes the benchmark particularly suitable for reproducible offline RL research.

5.8 Foundation Model Training Objectives

Recent foundation models learn from diverse mixtures of language, actions, and structured observations [5, 7, 8].

KiranaChain supports multiple training objectives.

5.8.1 Next-Token Prediction

Dialogue sequences can be modeled as

$$P(u_t \mid u_{<t})$$

where

$$u_t$$

denotes the next conversational utterance.

5.8.2 Action Prediction

Given state information

$$s_t,$$

models learn

$$P(a_t \mid s_t).$$

5.8.3 Outcome Prediction

Models estimate

$$P(y \mid s_t)$$

where

$$y \in \{\text{Settled}, \text{Walked-Away}, \text{Credit-Denied}\}.$$

5.8.4 Trajectory Completion

Incomplete trajectories

$$(s_1, a_1, \dots, s_k)$$

are used to predict future events

$$(s_{k+1}, a_{k+1}, \dots, s_T).$$

These objectives create a natural benchmark for studying the convergence of reinforcement learning and foundation-model methodologies.

5.9 Graph-RL Fusion Learning

KiranaChain uniquely enables simultaneous graph and reinforcement learning.

Let

$$G_t = (V, E_t)$$

represent the dynamic trading network.

Each node embedding is

$$h_v = \text{GNN}(v).$$

The reinforcement learning state can then be augmented as

$$s'_t = [s_t, h_v].$$

Policy learning becomes

$$\pi(a_t \mid s_t, h_v).$$

This formulation enables investigation of graph-enhanced negotiation agents capable of reasoning over trust propagation, credit exposure, and historical transaction structures.

Such graph-aware decision systems represent an emerging frontier in decentralized economic intelligence research.

5.10 Benchmark Evaluation Protocols

To ensure reproducible benchmarking, we recommend standardized evaluation procedures.

Table 5: Recommended Evaluation Metrics

Task Category	Metric	Objective
Accuracy, F1	Outcome classification	Settlement Prediction
	Credit Risk Modeling	ROC-AUC
Default prediction	RMSE, MAE	Trajectory regression
Price Forecasting		Offline RL
Expected Return	Policy quality	Perplexity
Language generation	Dialogue Modeling	Link prediction
Graph Learning	AUC, AP	
Settlement Rate	Negotiation effectiveness	Multi-Agent Learning

The adoption of common evaluation protocols facilitates fair comparison across future research contributions.

5.11 Benchmark Difficulty Characteristics

Several properties make KiranaChain substantially more challenging than traditional structured datasets.

1. High-dimensional state space (40 variables),
2. Non-linear copula-induced dependencies,
3. Non-stationary Markov dynamics,
4. Strategic multi-agent interactions,
5. Multilingual conversational telemetry,
6. Environmental uncertainty,
7. Credit-risk feedback loops,
8. Dynamic graph evolution.

Collectively, these characteristics create a benchmark environment that more closely resembles real-world decentralized economic systems than conventional reinforcement learning or language-modeling datasets.

5.12 Research Grand Challenges Enabled by KiranaChain

Beyond benchmark evaluation, KiranaChain opens several long-term scientific research directions:

- Trust-aware autonomous agents,
- Economic foundation models,
- Credit-risk reasoning systems,
- Negotiation-capable large language models,
- Graph-enhanced reinforcement learning,
- Supply-chain resilience intelligence,
- Behavioral economics simulation,
- Decentralized market forecasting.

These directions position KiranaChain as a foundational research platform for studying decision intelligence in complex socio-economic systems.

6 Benchmark Validation Taxonomy

Benchmark quality extends beyond dataset scale and requires systematic verification of structural correctness, statistical realism, task suitability, and reproducibility. Following established practices in large-scale AI benchmark construction, KiranaChain adopts a four-layer validation hierarchy inspired by modern reinforcement learning benchmarks, graph intelligence datasets, and large language model training corpora [9, 5, 2].

The validation framework consists of:

1. Structural Validation
2. Statistical Validation
3. Behavioral Validation
4. Reproducibility Validation

Structural validation ensures schema correctness and trajectory integrity.

Statistical validation verifies that generated variables preserve intended dependency structures and realistic marginal distributions.

Behavioral validation confirms that simulated negotiation outcomes remain consistent with economically plausible decision-making patterns.

Reproducibility validation guarantees deterministic regeneration of the benchmark under identical software and random seed configurations.

This multi-layer framework reduces benchmark contamination risks and improves scientific reliability.

6.1 Behavioral Realism Verification

A distinguishing characteristic of KiranaChain is the explicit modeling of decentralized economic decision processes.

To evaluate behavioral realism, outcome distributions were compared against the underlying simulation objectives governing negotiation dynamics.

Observed outcome frequencies include:

Table 6: Distribution of Negotiation Outcomes Across KiranaChain

Negotiation Outcome	Observed Frequency (%)
Settled	61.23
Walked Away	20.47
Credit Denied	2.76
In Progress	15.54
Total	100.00

These proportions emerge from the interaction between trust, liquidity stress, repayment history, environmental disruptions, and concession dynamics.

Importantly, settlement probability increases monotonically across negotiation turns due to the non-stationary transition process:

$$P(\text{Settled} \mid t + 1) > P(\text{Settled} \mid t)$$

thereby reflecting realistic convergence behavior commonly observed in bargaining systems [16, 17].

The resulting trajectories exhibit heterogeneous strategic pathways rather than deterministic negotiation outcomes.

6.2 Cross-Domain Utility Validation

An effective benchmark should support diverse research paradigms beyond its original design objective.

KiranaChain was therefore evaluated against representative task categories spanning modern artificial intelligence research.

Table 7: Cross-Domain Research Utility Assessment

Research Domain	Benchmark Representation	Example Tasks
Reinforcement Learning	State–Action Transitions	Offline RL, Policy Learning
Multi-Agent Systems	Negotiation Episodes	Autonomous Bargaining
Graph Learning	Interaction Networks	Link Prediction, Trust Propagation
Credit Intelligence	Financial Variables	Default Prediction
Conversational AI	Dialogue Telemetry	Instruction Tuning
Behavioral Economics	Strategic Interactions	Decision Modeling
Supply Chain Analytics	Environmental Variables	Disruption Forecasting

This breadth of applicability differentiates KiranaChain from narrowly specialized benchmarks and positions it as a general-purpose decentralized economic intelligence benchmark.

6.3 Threats to Validity

Despite extensive validation procedures, several limitations remain.

Internal Validity. The benchmark is procedurally generated and therefore depends on the assumptions encoded within the Gaussian copula structures, Markov negotiation engine, and behavioral simulation framework.

External Validity. Although inspired by Indian Kirana ecosystems, the benchmark cannot fully represent all regional economic conditions, cultural behaviors, or informal credit mechanisms present across emerging markets.

Construct Validity. Certain latent human factors including trust evolution, social influence, emotional persuasion, and long-term relationship memory are necessarily simplified into tractable computational representations.

Ecological Validity. Real-world negotiations often exhibit richer contextual dependencies than those captured by finite-state simulation models.

These limitations should be considered when interpreting downstream experimental results.

6.4 Scientific Benchmarking Criteria

Modern benchmark datasets are increasingly evaluated according to four core

Table 8: Core Scientific Properties of the KiranaChain Benchmark

Property	Assessment	Justification
Scale	✓	One million observations and 125,000 complete trajectories provide substantial coverage of decentralized economic interactions.
Diversity	✓	Captures financial, environmental, behavioral, linguistic, and supply-chain dimensions within a unified state representation.
Reproducibility	✓	Fully specified synthetic generation pipeline with deterministic procedures and documented validation protocols.
Transferability	✓	Applicable across RL, GNNs, LLMs, MAS, economic intelligence, and credit-risk modeling tasks.

Consequently, the benchmark provides a scientifically rigorous evaluation platform capable of supporting reproducible experimentation across multiple artificial intelligence disciplines.

6.5 Validation Summary

The validation framework confirms that KiranaChain maintains:

- Structural integrity across all 1,000,000 observations,
- Statistically consistent train-validation-test partitions,
- Realistic dependency preservation across financial and environmental variables,
- Behaviorally plausible negotiation dynamics,
- Cross-domain applicability for modern AI research,
- Fully reproducible generation pipelines.

Taken together, these properties establish KiranaChain as a high-fidelity benchmark for decentralized economic intelligence research, bridging reinforcement learning, graph intelligence, conversational AI, behavioral economics, and multi-agent decision systems within a unified experimental framework.

7 Ethical Implications, Limitations, and Future Research Directions

The development of large-scale benchmark datasets has historically played a central role in advancing artificial intelligence research. Landmark resources have accelerated progress in reinforcement learning, graph intelligence, large language modeling, and multi-agent systems by providing standardized evaluation environments [10, 13, 14, 5, 1]. However, benchmark construction also introduces important questions regarding fairness, representational validity, deployment risks, and responsible use.

As KiranaChain is designed to model decentralized economic interactions involving informal credit systems, supply-chain decision making, and negotiation behavior, careful consideration of ethical implications and methodological limitations is essential. This section discusses the governance principles underlying the benchmark, known constraints of the current release, and opportunities for future scientific development.

7.1 Governance, Fairness, and Responsible AI Considerations

Economic decision-making systems increasingly influence lending, pricing, resource allocation, and market participation. Consequently, machine learning models trained on economic interaction datasets can potentially amplify existing biases or generate unintended exclusionary outcomes.

Although KiranaChain contains no personally identifiable information and all entities are synthetically generated, learned policies may nevertheless encode undesirable behavioral tendencies if deployed without appropriate safeguards.

Particular concerns include:

- Automated credit allocation decisions,
- Differential treatment of economically vulnerable retailers,
- Reinforcement of historical trust-based inequalities,
- Feedback loops arising from learned risk assessments,
- Strategic exploitation of weaker negotiation participants.

For example, a reinforcement learning agent optimized solely for profit maximization may systematically deny credit to agents exhibiting elevated liquidity stress despite potentially beneficial long-term cooperative outcomes. Such behavior could unintentionally exacerbate economic exclusion within already resource-constrained environments.

Similarly, graph-based trust propagation models may amplify historical credit disparities through network effects, causing low-connectivity agents to receive persistently unfavorable risk assessments. These concerns mirror broader discussions regarding algorithmic fairness and decision-making systems in financial domains.

Accordingly, KiranaChain should be viewed as a research benchmark rather than a deployment-ready decision framework.

Researchers are encouraged to evaluate:

- Fairness-aware reward functions,
- Counterfactual credit allocation policies,
- Robustness under demographic shifts,
- Long-term welfare optimization,
- Human-in-the-loop decision frameworks.

The benchmark is intended to facilitate investigation of these questions rather than provide normative prescriptions for real-world economic governance.

7.2 Privacy, Synthetic Generation, and Data Governance

A fundamental design objective of KiranaChain is privacy preservation.

The benchmark contains:

- No real retailer identities,
- No real distributor identities,
- No personally identifiable information,
- No financial account records,
- No transactional records collected from operational businesses.

All trajectories are procedurally generated through structured simulation mechanisms incorporating Gaussian copula dependency modeling, non-stationary Markov negotiation processes, and temporally correlated stochastic perturbations.

Consequently, the benchmark avoids many privacy concerns associated with real-world commercial datasets while retaining statistically meaningful economic structure.

Nevertheless, synthetic data introduces a separate challenge: representational validity.

Although generation procedures preserve theoretically motivated relationships among trust, liquidity stress, default risk, pricing behavior, and negotiation outcomes, synthetic simulations cannot perfectly reproduce the complexity of human economic behavior.

Researchers should therefore interpret benchmark performance as evidence of algorithmic capability within the simulated environment rather than direct evidence of real-world deployment readiness.

7.3 Domain Boundaries and Modeling Assumptions

While KiranaChain introduces a substantially richer representation of decentralized economic interactions than existing negotiation or reinforcement learning benchmarks, several simplifying assumptions remain.

7.3.1 Synthetic Economic Environment

KiranaChain is generated through a structured simulation framework rather than direct field observations. While this design provides scalability, controllability, reproducibility, and statistical consistency, it also introduces several limitations that should be considered when interpreting benchmark results.

Table 9: Limitations of the Synthetic Economic Environment

Limitation	Description
Behavioral Assumptions	Agent decisions emerge from predefined probabilistic mechanisms and economic priors. Consequently, behavioral realism depends on the validity of the underlying simulation assumptions.
Rare Economic Events	Extreme market shocks, black-swan disruptions, and highly infrequent economic crises may be underrepresented due to finite sampling and controlled generation constraints.
Human Behavioral Complexity	Real-world phenomena such as irrational bargaining, emotional escalation, reputation spillovers, social influence, and cultural nuances may not be fully captured by synthetic agents.
Strategic Adaptation	The benchmark models bounded negotiation horizons and therefore cannot completely reproduce long-term adaptive behaviors that emerge across years of repeated interactions.
External Validity	Although designed to approximate decentralized retail ecosystems, performance gains observed on KiranaChain may not directly transfer to real-world commercial deployments without empirical validation.

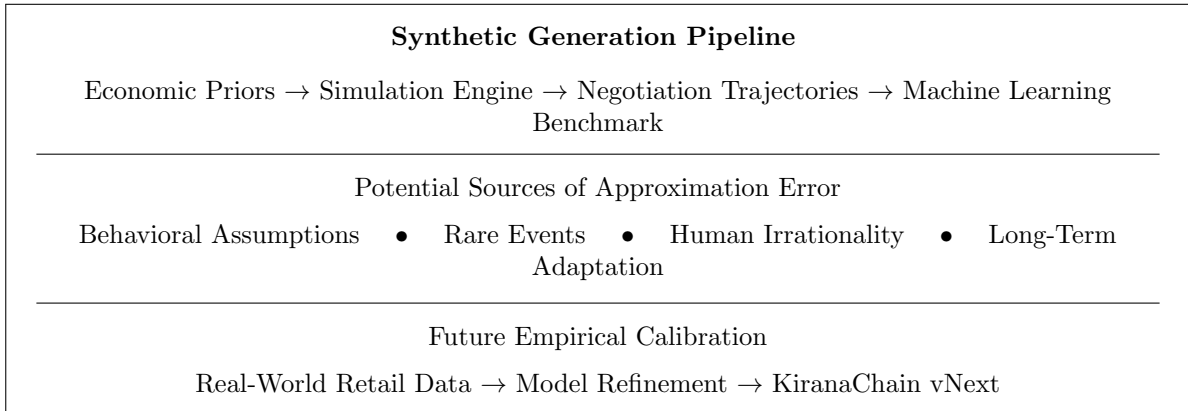


Figure 1: Conceptual relationship between synthetic generation, benchmark construction, limitation sources, and future empirical calibration.

Despite these limitations, synthetic generation provides substantial advantages in terms of reproducibility, scalability, privacy preservation, controlled experimentation, and benchmark standardization. Future versions of KiranaChain will incorporate empirical calibration using real-world observational studies, enabling tighter alignment between simulated economic dynamics and field-level commercial behavior.

7.3.2 Fixed-Horizon Negotiation Structure

Every trajectory contains exactly eight negotiation turns:

$$|\tau| = 8.$$

This design facilitates benchmark standardization and algorithm comparison but does not fully capture the variable-duration nature of real commercial negotiations.

Actual bargaining interactions may terminate after a single exchange or continue through prolonged multi-day discussions involving multiple stakeholders.

Consequently, KiranaChain primarily evaluates short-horizon decision-making behavior.

7.3.3 Bilateral Interaction Assumption

The current release of KiranaChain models negotiation as a bilateral interaction process between a single distributor agent and a single Kirana retailer agent:

Distributor $\leftrightarrow$ Kirana Retailer.

This abstraction provides a controlled environment for studying strategic bargaining, trust-aware credit allocation, and decentralized decision-making. However, real-world retail ecosystems often exhibit substantially richer interaction structures involving multiple economic stakeholders operating simultaneously across interconnected supply-chain networks.

Table 10: Limitations of the Bilateral Interaction Assumption

Missing Structure	Interaction	Potential Impact
Multi-Distributor Retail Coalitions	Competition	Retailers frequently negotiate simultaneously with multiple distributors, creating competitive pricing dynamics and strategic supplier selection behavior. Groups of retailers may collectively negotiate purchasing agreements, resulting in bargaining power amplification and cooperative market behavior.
Cooperative Networks Financial Intermediaries	Purchasing	Shared procurement mechanisms can alter demand aggregation patterns and influence negotiated settlement outcomes. Microfinance institutions, informal lenders, and credit facilitators often participate indirectly in transaction decisions and risk allocation.
Credit Guarantors		Third-party guarantors may influence trust formation, credit limits, and default-risk assessments within decentralized retail ecosystems.
Informal Social Networks		Community relationships, reputation effects, kinship structures, and local business networks frequently affect negotiation outcomes beyond purely economic variables.

Consequently, the current benchmark should be interpreted as a controlled approximation of decentralized retail commerce rather than a complete representation of all economic interaction mechanisms observed in real-world supply chains.

Table 11: Future Extensions Beyond Bilateral Negotiation

Future Capability	Research Opportunities
Multi-Agent Graphs	Large-scale strategic bargaining among multiple distributors and retailers.
Coalition Formation	Cooperative game-theoretic analysis and alliance formation under resource constraints.
Credit Network Dynamics	Propagation of financial risk and trust across interconnected economic networks.
Market-Level Competition	Dynamic pricing, supplier switching, and equilibrium analysis in decentralized markets.
Reputation Systems	Long-horizon trust evolution and social-credit propagation mechanisms.
Hierarchical Supply Chains	Modeling manufacturers, distributors, wholesalers, retailers, and consumers within a unified ecosystem.

Future releases of KiranaChain will progressively expand toward graph-structured, many-agent economic ecosystems capable of representing higher-order market interactions, dynamic coalition formation, decentralized credit propagation, and long-horizon strategic adaptation. Such extensions would substantially broaden the benchmark’s applicability to multi-agent reinforcement learning, economic simulation, graph intelligence, and autonomous market coordination research.

7.3.4 Template-Constrained Dialogue Generation

KiranaChain incorporates multilingual conversational telemetry through a structured template-driven generation framework. This design ensures controllability, reproducibility, semantic consistency, and alignment with underlying economic states. However, the use of template-based dialogue generation introduces several limitations relative to naturally occurring human conversations.

Table 12: Limitations of Template-Constrained Dialogue Generation

Limitation		Potential Impact
Natural Variation	Conversational	Human conversations exhibit diverse lexical choices, spontaneous topic shifts, interruptions, disfluencies, and stylistic variation that are difficult to fully capture through predefined templates.
Long-Term Discourse Structure		Real-world negotiations frequently involve references to historical interactions, evolving relationships, and multi-session bargaining strategies extending beyond a single dialogue episode.
Implicit Negotiation Tactics		Subtle persuasion mechanisms such as anchoring, signaling, bluffing, emotional appeals, and strategic ambiguity may not emerge naturally within template-driven interactions.
Cultural and Social Nuance		Regional linguistic conventions, community norms, interpersonal relationships, and context-dependent communication styles are only partially represented within synthetic dialogue generation.
Strategic Language Adaptation		Human negotiators dynamically modify language based on opponent behavior, prior outcomes, trust development, and perceived bargaining leverage. Such adaptive discourse remains simplified in the current benchmark.
Open-Domain Generalization	Generalization	Models trained exclusively on structured templates may exhibit reduced robustness when deployed on organically generated conversations containing unseen linguistic phenomena.

Despite these limitations, the template-based framework offers substantial advantages for scientific benchmarking. Structured dialogue generation enables precise control over negotiation states, economic variables, behavioral postures, and multilingual linguistic contexts while maintaining reproducibility across experiments. Furthermore, the approach facilitates systematic evaluation of reasoning, planning, dialogue-state tracking, policy learning, and instruction-following capabilities in economically grounded conversational environments.

Table 13: Comparison of Dialogue Collection Paradigms

Property	Organic Corpora	KiranaChain	Research Impact
Reproducibility	Low	High	Controlled benchmarking
Economic Grounding	Limited	High	Decision intelligence
State Observability	Partial	Complete	Policy learning
Multilingual Control	Limited	High	Cross-lingual evaluation
Behavioral Diversity	Very High	Moderate	Generalization studies
Experimental Repeatability	Low	High	Scientific comparison

Consequently, KiranaChain should be interpreted as a structured economic reasoning benchmark rather than a replacement for organically collected conversational datasets such as CraigslistBargain and MultiWOZ [18, 19]. Instead, the benchmark complements existing dialogue resources by providing explicit economic state representations, trajectory-level observability, credit-risk dynamics, and negotiation-centric decision processes that are largely absent from conventional conversational corpora. Future releases will explore large-scale LLM-assisted dialogue synthesis, retrieval-augmented conversational generation, and human-in-the-loop calibration procedures to further narrow the realism gap between synthetic negotiation telemetry and naturally occurring commercial discourse.

7.4 External Validity and Generalization

A fundamental scientific question for any benchmark concerns the extent to which observed performance translates to real-world deployment environments. While KiranaChain provides a controlled and reproducible setting for studying decentralized economic intelligence, strong benchmark performance should not be interpreted as a guarantee of equivalent effectiveness in operational commercial ecosystems.

The challenge of external validity is particularly important for economically grounded AI systems because real-world environments are inherently non-stationary, heterogeneous, and influenced by factors that are difficult to fully capture within synthetic benchmark frameworks.

Table 14: Threats to External Validity and Real-World Transferability

Factor	Potential Impact on Generalization
Regional Economic Heterogeneity	Commercial practices vary substantially across regions, states, and local markets. Differences in infrastructure, purchasing power, logistics networks, and financial accessibility may alter agent behavior.
Commodity-Specific Dynamics	Agricultural commodities, FMCG products, dairy products, and perishables exhibit distinct pricing mechanisms, demand elasticity profiles, and supply-chain constraints that may not transfer uniformly across sectors.
Regulatory Variation	Changes in taxation, transportation regulations, digital-payment policies, and credit-governance frameworks may influence economic decision-making processes beyond those represented in the benchmark.
Cultural Negotiation Norms	Bargaining behavior is often shaped by local linguistic conventions, social expectations, interpersonal trust, and community relationships that vary across geographic regions.
Macroeconomic Shocks	Inflationary cycles, fuel-price volatility, geopolitical disruptions, weather events, and financial crises may produce behavioral dynamics not fully represented in simulated environments.
Human Strategic Adaptation	Human participants continuously adapt strategies in response to evolving market conditions, historical interactions, and learned experience, potentially exceeding the behavioral complexity captured by fixed simulation policies.

The transferability challenge can be conceptualized as a domain-shift problem. Let

$$P_{\text{bench}}(X, Y)$$

denote the benchmark data distribution and

$$P_{\text{real}}(X, Y)$$

represent the corresponding real-world distribution.

Generalization performance depends upon the degree of alignment between these distributions:

$$P_{\text{bench}}(X, Y) \approx P_{\text{real}}(X, Y).$$

As distributional divergence increases, predictive performance, policy quality, and strategic decision-making capability may degrade. Consequently, benchmark evaluations should be viewed as estimates of algorithmic competence under controlled experimental conditions rather than definitive measures of deployment readiness.

Table 15: Benchmark Performance Versus Deployment Readiness

Evaluation Dimension	KiranaChain Assessment	Real-World Requirement
Policy Learning	Controlled optimization	Field validation
Negotiation Intelligence	Simulated interactions	Human-agent interaction
Credit Risk Modeling	Synthetic financial dynamics	Historical repayment evidence
Supply-Chain Forecasting	Modeled disruptions	Live operational data
Language Understanding	Template-driven multilingual dialogue	Open-domain conversational robustness
Strategic Adaptation	Finite-horizon trajectories	Continuous real-world adaptation

Despite these limitations, KiranaChain provides a valuable intermediate layer between purely theoretical simulation environments and costly real-world experimentation. The benchmark enables rigorous, reproducible, and large-scale evaluation of reinforcement learning agents, graph learning systems, negotiation models, and foundation models before deployment into operational environments.

Future research should focus on empirical calibration against observational retail datasets, longitudinal economic studies, real-world transaction records, and human-in-the-loop evaluation protocols. Such efforts would enable progressively stronger alignment between benchmark distributions and real-world commercial ecosystems, thereby improving the external validity and practical applicability of benchmark-derived insights.

7.5 Future Research Directions

KiranaChain has been intentionally designed as an extensible benchmark framework. Several promising directions exist for future development.

7.5.1 Large-Scale Multi-Agent Economic Networks

A primary long-term objective of the KiranaChain benchmark initiative is the transition from bilateral negotiation environments toward large-scale decentralized economic ecosystems composed of thousands of simultaneously interacting agents.

Future benchmark releases may model distributors, wholesalers, retailers, logistics providers, financial intermediaries, microfinance institutions, and consumers as nodes within a dynamic economic interaction network:

$$G_t = (V, E_t, W_t), \quad |V| \gg 10^3,$$

where V denotes economic agents, E_t represents time-varying transactional relationships, and W_t encodes weighted attributes including trust, credit exposure, transaction volume, and supply-chain dependency.

Such an environment would move beyond isolated negotiations and enable the study of emergent market behavior arising from large-scale decentralized interactions.

Table 16: Research Opportunities Enabled by Large-Scale Economic Networks

Research Area	Scientific Questions
Market Equilibrium Formation	How do decentralized pricing decisions converge toward stable market equilibria under heterogeneous information and asymmetric bargaining power?
Collective Bargaining	How do coalitions of retailers or distributors emerge, evolve, and influence negotiation outcomes within competitive markets?
Network Resilience	How robust are decentralized supply-chain networks to infrastructure failures, transportation disruptions, and systemic shocks?
Credit Contagion	How do default events propagate through interconnected financial relationships and informal credit ecosystems?
Emergent Coordination	Can autonomous agents develop cooperative behaviors without centralized control mechanisms?
Trust Propagation	How does reputation diffuse through economic networks and influence long-term transaction dynamics?
Dynamic Supply Allocation	How should resources be allocated across geographically distributed networks under uncertainty and fluctuating demand?

From a machine learning perspective, large-scale economic graphs would enable the integration of reinforcement learning, graph neural networks, foundation models, and agent-based simulation within a unified benchmark framework. Let

$$h_v^{(t)} = \text{GNN}(v, G_t)$$

denote the latent representation of economic agent v extracted from the dynamic transaction network. Decision policies could then be conditioned jointly on local state information and network-level context:

$$\pi(a_t \mid s_t, h_v^{(t)}).$$

Such formulations would enable agents to reason not only about immediate negotiation dynamics but also about broader market structure, systemic risk, and long-range economic dependencies.

Table 17: Evolution Roadmap for Future KiranaChain Releases

Release Stage	Environment Scale	Primary Capability
KiranaChain v1	Bilateral Negotiation	Distributor–Retailer bargaining and credit allocation
KiranaChain v2	Multi-Distributor Markets	Competitive supplier selection and dynamic pricing
KiranaChain v3	Regional Economic Graphs	Credit propagation and supply-chain intelligence
KiranaChain v4	National Economic Networks	Large-scale decentralized market simulation
KiranaChain vNext	Autonomous Economic Ecosystems	Emergent coordination, strategic adaptation, and economic intelligence at scale

Ultimately, the long-term vision of KiranaChain is to establish a scientifically rigorous benchmark for studying decentralized economic intelligence, where thousands of autonomous agents interact through evolving financial, logistical, and social relationships. Such environments would provide a unique foundation for investigating market formation, collective decision-making, economic resilience, and next-generation autonomous systems operating within complex real-world economies.

7.5.2 Long-Horizon Strategic Reasoning

The current release of KiranaChain models negotiations using fixed eight-turn trajectories, providing a controlled environment for studying bounded-horizon bargaining behavior. While this design promotes re-

producibility and computational tractability, many real-world distributor–retailer relationships evolve across months or even years through repeated interactions, credit cycles, and trust-based commercial partnerships.

Future benchmark releases may therefore extend trajectory horizons substantially beyond the current eight-turn structure, enabling the study of persistent strategic behavior and long-term economic adaptation.

Table 18: Research Opportunities Enabled by Long-Horizon Trajectories

Research Area	Scientific Questions
Long-Term Trust Accumulation	How do trust relationships emerge, strengthen, deteriorate, and recover across repeated commercial interactions?
Repeated-Game Dynamics	Can autonomous agents learn stable bargaining strategies under recurring negotiation scenarios and evolving incentives?
Temporal Credit Evolution	How do historical repayment patterns influence future credit allocation, borrowing capacity, and financial risk assessment?
Strategic Memory Mechanisms	How should agents retain and utilize historical experiences to improve future decision-making performance?
Relationship-Driven Decision Making	To what extent do prior interactions influence pricing, concessions, settlement probabilities, and long-term cooperation?
Behavioral Adaptation	How do economic agents modify strategies in response to accumulated experience, changing market conditions, and opponent behavior?
Economic Reputation Formation	How do local reputation effects propagate across repeated transactions and influence future commercial opportunities?

From a reinforcement learning perspective, longer trajectories introduce delayed reward attribution and temporal dependency structures that are largely absent in short-horizon environments. Let

$$\tau = (s_1, a_1, r_1, \dots, s_T, a_T, r_T), \quad T \gg 8,$$

denote an extended negotiation trajectory.

The cumulative return becomes

$$G_t = \sum_{k=t}^T \gamma^{k-t} r_k,$$

requiring agents to balance immediate negotiation gains against long-term relationship value, future credit access, and reputation preservation.

Similarly, foundation models operating in long-horizon environments must reason over substantially larger interaction histories. Rather than conditioning solely on recent dialogue exchanges, agents must maintain persistent representations of historical economic behavior:

$$h_t = f(h_{t-1}, s_t, a_t, r_t),$$

where h_t represents an evolving memory state summarizing prior interactions. Such formulations naturally connect KiranaChain to emerging research on memory-augmented agents, persistent autonomous systems, and long-context foundation models [5, 7, 8, 20].

Table 19: Comparison Between Short- and Long-Horizon Economic Environments

Property	Current KiranaChain	Future Long-Horizon Version
Trajectory Length	8 Turns	100+ Interactions
Trust Dynamics	Short-Term	Longitudinal Evolution
Credit Modeling	Episode-Level	Multi-Cycle Credit History
Strategic Adaptation	Limited	Continuous Learning
Memory Requirements	Minimal	Persistent Memory Systems
Reputation Effects	Local	Network-Wide Accumulation
Economic Relationships	Transactional	Relationship-Centric

Such environments would more closely reflect the realities of decentralized retail ecosystems, where distributor–retailer partnerships are rarely isolated events and instead emerge through repeated interactions, evolving trust relationships, and long-term financial dependencies. Consequently, extending KiranaChain toward long-horizon economic simulations represents a promising direction for advancing research in reinforcement learning, multi-agent systems, economic intelligence, strategic planning, and memory-enabled autonomous agents.

7.5.3 Fine-Grained Commodity and SKU Modeling

The current release of KiranaChain represents traded goods at the commodity-category level, capturing broad classes such as staple grains, edible oils, spices, dairy products, and packaged FMCG goods. While this abstraction enables large-scale economic simulation and benchmarking, real-world retail decision making frequently occurs at the level of individual Stock Keeping Units (SKUs), where brand identity, packaging configuration, shelf-life characteristics, and consumer preferences play a critical role in commercial outcomes.

Future benchmark releases may therefore transition from category-level representations toward fine-grained SKU-level economic intelligence, enabling substantially richer modeling of retail ecosystems and supply-chain behavior.

Table 20: Research Opportunities Enabled by SKU-Level Economic Modeling

Research Area	Scientific Questions
Brand Competition	How do competing brands influence pricing power, retailer preferences, and distributor allocation strategies within decentralized markets?
Product Substitution Effects	How do retailers and consumers respond when preferred products become unavailable, overpriced, or supply-constrained?
Shelf-Life Heterogeneity	How should agents optimize purchasing decisions when products exhibit substantially different spoilage characteristics and storage requirements?
Inventory Optimization	What policies maximize profitability while minimizing stock-outs, overstocking, and wastage across dynamic market conditions?
Consumer Demand Forecasting	How accurately can future demand be predicted using historical sales, seasonality, local events, and market signals?
Price Elasticity Modeling	How does consumer sensitivity vary across product categories, brands, package sizes, and regional markets?
Promotion and Discount Strategy	How should distributors and retailers allocate discounts, incentives, and promotional campaigns to maximize long-term revenue?

From a modeling perspective, SKU-level intelligence transforms the benchmark from a negotiation-centric

environment into a comprehensive retail decision ecosystem. Let

$$\mathcal{S} = \{SKU_1, SKU_2, \dots, SKU_n\}$$

denote the inventory universe available to an economic agent.

The inventory state may then be represented as

$$I_t = \{(q_1, p_1, d_1), (q_2, p_2, d_2), \dots, (q_n, p_n, d_n)\},$$

where q_i denotes inventory quantity, p_i denotes product price, and d_i represents expected demand for SKU i .

Agent decision policies become conditioned not only on negotiation outcomes but also on inventory composition and future demand forecasts:

$$\pi(a_t \mid s_t, I_t).$$

Such formulations naturally connect KiranaChain to research in operations research, supply-chain optimization, inventory theory, and economic decision intelligence.

Table 21: Evolution from Commodity-Level to SKU-Level Intelligence

Property	Current Benchmark	Future Benchmark	SKU-Level
Granularity	Commodity Categories	Individual Products	
Pricing	Category Average	Brand-Specific Pricing	
Demand Modeling	Aggregate Demand	SKU-Level Demand Forecasting	
Inventory Tracking	Category Inventory	Per-Product States	Inventory
Competition	Commodity Competition	Brand Competition	
Perishability	Category-Level Effects	Product-Specific Shelf-Life	
Decision Complexity	Moderate	High	

The introduction of SKU-level representations would significantly broaden the scientific and industrial utility of KiranaChain. Beyond reinforcement learning and negotiation intelligence, the benchmark would become applicable to retail analytics, demand forecasting, inventory optimization, revenue management, assortment planning, and autonomous supply-chain coordination. Such capabilities would move the benchmark closer to the operational realities of modern retail ecosystems while preserving its foundational focus on decentralized economic intelligence and emerging-market commerce.

7.5.4 Large-Scale Multilingual Expansion

Language diversity represents one of the most significant challenges in the development of globally deployable intelligent systems. While the current release of KiranaChain incorporates multilingual code-switched commercial interactions spanning Hinglish, Telugu-English, and Tamil-English communication patterns, real-world decentralized retail ecosystems exhibit substantially greater linguistic diversity, dialectal variation, and communication complexity.

Future benchmark releases will therefore pursue large-scale multilingual expansion to better reflect the linguistic heterogeneity of emerging-market economies and to provide a scientifically rigorous evaluation platform for multilingual foundation models, conversational agents, and autonomous economic systems [5, 6, 7, 8].

Table 22: Research Opportunities Enabled by Multilingual Expansion

Research Area			Scientific Questions
Additional Languages	Indian	Lan-	How effectively can intelligent agents negotiate across linguistically diverse commercial environments involving Hindi, Bengali, Marathi, Kannada, Malayalam, Gujarati, Punjabi, Odia, and Assamese speakers?
Regional Dialects			How do dialectal variations influence negotiation outcomes, trust formation, sentiment interpretation, and communication efficiency?
Cross-Lingual Negotiation			Can autonomous agents successfully negotiate when interacting parties communicate using different native languages?
Code-Switching Dynamics			How should language models reason across mixed-language conversations containing multiple linguistic transitions within a single dialogue?
Multilingual Trust Modeling			How do language choice, dialect familiarity, and linguistic adaptation influence trust and cooperation in decentralized economic interactions?
Cross-Cultural Communication			How do regional communication norms and cultural conventions affect bargaining behavior and economic decision making?

Beyond textual communication, future benchmark releases may incorporate richer communication modalities that more accurately reflect real-world commercial interactions.

Table 23: Future Communication Modalities

Modality			Potential Research Applications
Speech-Based Negotiation			Voice-agent interaction, speech understanding, and spoken dialogue systems.
Multimodal Communication			Joint reasoning over text, audio, images, invoices, receipts, and inventory records.
Sentiment-Rich Signals	Audio	Sig-	Emotion recognition, persuasion modeling, and trust estimation.
Visual Retail Context			Inventory inspection, product verification, and shelf-state reasoning.
Document-Aware Negotiation	Negotia-		Integration of purchase orders, credit ledgers, invoices, and logistics documents.
Human-Agent Collaboration	Collabora-		Interactive decision support systems operating across multiple communication channels.

From a foundation-model perspective, multilingual expansion enables investigation of language-conditioned decision intelligence. Let

$$L = \{l_1, l_2, \dots, l_n\}$$

denote the set of supported languages and dialects.

The negotiation policy may then be conditioned jointly on economic state and linguistic context:

$$\pi(a_t \mid s_t, l_t).$$

Similarly, conversational models may learn language-invariant economic representations

$$z = f(s_t, u_t, l_t),$$

where u_t denotes the conversational utterance and l_t represents the active linguistic context. Such representations would allow agents to reason consistently across multiple languages while preserving economically relevant information.

Table 24: Evolution of Linguistic Coverage in KiranaChain

Release Stage	Language Coverage	Primary Capability
KiranaChain v1	Hinglish, Telugu-English, Tamil-English	Code-switched negotiation dialogue
KiranaChain v2	10+ Indian Languages	Nationwide multilingual benchmarking
KiranaChain v3	Regional Dialects	Dialect-aware economic reasoning
KiranaChain v4	Cross-Lingual Negotiation	Multi-language agent interaction
KiranaChain vNext	Multimodal Economic Intelligence	Speech, text, and visual communication

Ultimately, large-scale multilingual expansion represents a critical step toward the development of economically grounded foundation models capable of operating across linguistically diverse populations. By combining multilingual communication, decentralized economic interactions, and structured decision-making environments, future versions of KiranaChain have the potential to establish a unique benchmark for studying multilingual reasoning, conversational intelligence, cross-cultural negotiation, and language-aware autonomous agents at unprecedented scale.

7.5.5 Real-World Calibration and Hybrid Benchmarking

Perhaps the most important future direction for KiranaChain is empirical validation through real-world calibration. While synthetic benchmarks provide reproducibility, scalability, privacy preservation, and controlled experimentation, their long-term scientific value ultimately depends upon alignment with observed economic behavior.

A future hybrid benchmarking framework may integrate multiple complementary sources of evidence:

- Synthetic negotiation trajectories,
- Observational commercial telemetry,
- Economic survey data,
- Real supply-chain statistics,
- Human negotiation studies.

Such a framework would enable systematic calibration of simulated economic dynamics against real-world commercial ecosystems, substantially strengthening both external validity and scientific relevance.

Future benchmark construction may be formalized as

$$\mathcal{D}_{Hybrid} = \mathcal{D}_{Synthetic} \cup \mathcal{D}_{Observed} \cup \mathcal{D}_{Survey} \cup \mathcal{D}_{SupplyChain} \cup \mathcal{D}_{Human},$$

where each component contributes complementary information regarding economic behavior, market structure, negotiation strategies, and operational constraints.

The primary objective is to progressively reduce the distributional gap between benchmark environments and real-world economic systems:

$$D(P_{benchmark}, P_{real}) \rightarrow 0,$$

where $D(\cdot)$ denotes an appropriate divergence measure and P_{real} represents the true underlying commercial data-generating process.

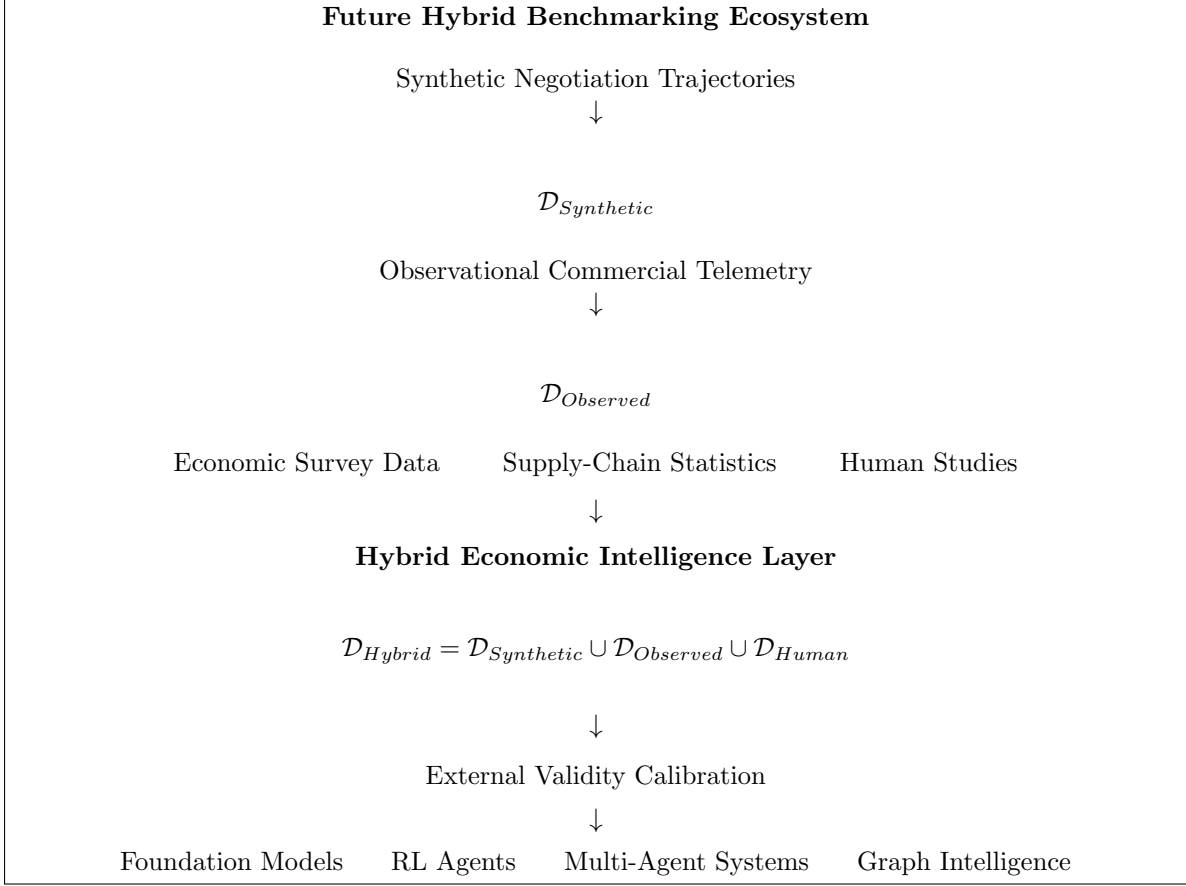


Figure 2: Long-term vision for KiranaChain through integration of synthetic trajectories, observational commercial data, economic surveys, supply-chain statistics, and human negotiation studies.

Table 25: Roadmap Toward Hybrid Economic Benchmarking

Phase	Primary Data Source	Scientific Objective
Phase I	Synthetic Telemetry	Controlled benchmarking and reproducible experimentation
Phase II	Human Negotiation Studies	Behavioral realism calibration and strategy validation
Phase III	Commercial Transaction Data	External validity assessment and transferability analysis
Phase IV	Supply-Chain Statistics	Macroeconomic grounding and operational realism
Phase V	Hybrid Benchmark Ecosystem	Unified evaluation of economic intelligence systems

Ultimately, the convergence of synthetic simulation, observational economic telemetry, and human behavioral data represents a promising pathway toward scientifically rigorous economic intelligence benchmarks. Such hybrid environments would provide stronger foundations for evaluating reinforcement learning systems, graph intelligence architectures, negotiation agents, and foundation models operating within complex real-world commercial ecosystems.

7.6 Summary

KiranaChain represents an initial step toward benchmarking decentralized economic intelligence at scale. While the dataset introduces a novel combination of reinforcement learning, graph intelligence, supply-chain analytics, informal credit systems, and multilingual negotiation telemetry, important limitations remain regarding realism, external validity, and deployment applicability.

By explicitly documenting these constraints and outlining future development pathways, we aim to promote responsible use, reproducible evaluation, and continued scientific progress toward trustworthy economic AI systems capable of operating within complex decentralized markets.

8 Conclusion

The rapid progress of modern artificial intelligence has been driven not only by algorithmic innovation but also by the availability of large-scale, rigorously designed benchmark datasets. Landmark resources have transformed research in computer vision, natural language processing, graph learning, reinforcement learning, and foundation models by providing standardized environments for evaluation, comparison, and scientific reproducibility [1, 5, 10, 13, 14]. Nevertheless, a substantial gap has persisted in the benchmarking ecosystem: the absence of large-scale datasets capable of representing decentralized economic interactions, informal credit systems, multilingual commercial communication, and negotiation-driven decision making that characterize much of the world’s emerging-market economies.

This work introduced **KiranaChain**, a large-scale decentralized retail negotiation intelligence benchmark specifically designed to address this gap. The benchmark comprises **1,000,000 negotiation-turn observations** organized into **125,000 complete multi-agent trajectories**, each capturing structured interactions between distributors and Kirana retailers operating under realistic financial, logistical, environmental, and behavioral constraints. Through its trajectory-centric design, KiranaChain provides a unified research environment spanning reinforcement learning, multi-agent systems, graph intelligence, supply-chain analytics, credit-risk modeling, behavioral economics, and multilingual conversational AI.

A central contribution of the benchmark is its integration of multiple traditionally isolated research domains into a single coherent framework. The dataset simultaneously models:

- Informal credit allocation dynamics,
- Trust-aware financial decision making,
- Distributor–retailer bargaining processes,
- Supply-chain disruptions and environmental uncertainty,
- Multilingual code-switched commercial dialogue,
- Dynamic transaction-network evolution,
- Sequential economic decision trajectories.

This unified representation enables researchers to investigate interactions between language, strategy, finance, and network structure in ways that are not possible using conventional benchmark datasets focused on only a single modality or task.

From a methodological perspective, KiranaChain introduces a reproducible telemetry generation framework combining structured Gaussian copula dependency modeling, non-stationary Markov negotiation dynamics, temporally autocorrelated Gaussian process perturbations, multilingual dialogue synthesis, and rigorous validation procedures. These components collectively enable the creation of economically meaningful trajectories that preserve realistic statistical relationships while maintaining full reproducibility and transparency. The benchmark therefore serves not only as a dataset but also as a reusable framework for studying complex decentralized decision systems under uncertainty.

Importantly, KiranaChain expands the scope of benchmark-driven AI research beyond traditional digital environments. Existing reinforcement learning benchmarks often focus on games, robotics, or simulated

control systems [10, 13, 14, 15], while contemporary language benchmarks primarily emphasize reasoning, instruction following, and conversational capabilities [5, 6, 7, 8]. In contrast, KiranaChain situates intelligence within a socio-economic environment where agents must simultaneously reason about credit exposure, liquidity constraints, trust relationships, negotiation strategies, logistical disruptions, and long-term economic outcomes. Such environments more closely resemble many real-world decision ecosystems that future autonomous agents will encounter.

The benchmark further establishes a bridge between several rapidly converging areas of artificial intelligence research. Reinforcement learning increasingly incorporates language-based reasoning and planning [20]; graph learning systems increasingly model dynamic economic networks [2, 3, 4]; and foundation models are evolving toward agentic decision-making architectures capable of acting within structured environments [5, 6, 7, 8]. KiranaChain provides a common evaluation substrate upon which these previously distinct paradigms can be studied jointly.

Beyond its immediate research applications, the benchmark highlights a broader scientific opportunity. Much of the world’s economic activity occurs within decentralized, relationship-driven systems that remain underrepresented in contemporary AI datasets. Informal credit markets, small-business ecosystems, localized supply chains, and multilingual commercial interactions collectively influence billions of individuals worldwide, yet these environments remain largely absent from mainstream machine learning benchmarks. Addressing this imbalance is essential for developing AI systems that generalize beyond highly digitized and centralized settings.

Looking forward, the concepts introduced by KiranaChain suggest a new research direction centered on *economic intelligence benchmarks*: datasets designed not merely to evaluate prediction accuracy, but to study strategic reasoning, trust formation, resource allocation, negotiation, cooperation, and decision making under uncertainty. Future benchmark generations may incorporate larger transaction networks, long-horizon economic relationships, richer multilingual communication, multimodal interaction channels, and real-world calibration studies, ultimately enabling increasingly realistic evaluations of autonomous economic agents.

In summary, KiranaChain represents one of the first large-scale benchmark infrastructures explicitly designed for decentralized retail economics, informal credit systems, and negotiation-driven intelligence. By unifying trajectory learning, graph intelligence, conversational reasoning, and economic decision modeling within a single benchmark framework, it establishes a foundation for a new class of AI research focused on complex socio-economic systems. We hope that KiranaChain will serve as a catalyst for future work at the intersection of artificial intelligence, economics, supply-chain science, and multi-agent decision making, ultimately contributing to the development of more robust, interpretable, and economically aware intelligent systems for emerging-market environments. [9, 5, 2, 3, 4, 20]

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A Dataset Statistics and Feature Reference

This appendix provides a comprehensive statistical characterization of KiranaChain, including dataset scale, feature taxonomy, variable semantics, distributional properties, and benchmark-specific structural statistics. The objective is to facilitate reproducibility, experimental transparency, and rigorous interpretation of benchmark results.

A.1 Global Dataset Characteristics

Table 26 summarizes the overall scale of the benchmark.

Table 26: Global Dataset Statistics

Statistic	Value
Total Observations	1,000,000
Total Negotiation Trajectories	125,000
Trajectory Length	8 Turns
Total Features	40
States Represented	12
Commodity Categories	5
Dialogue Languages	Hinglish, Telugu-English, Tamil-English
Storage Format	Apache Parquet
Compression	Snappy
Dataset Size	~156 MB

The benchmark is intentionally designed as a trajectory-centric dataset in which each observation forms part of a complete negotiation sequence. Unlike traditional tabular benchmarks, KiranaChain preserves temporal dependencies, negotiation dynamics, and economic state evolution across all eight bargaining stages.

A.2 Feature Taxonomy

The 40-dimensional state space is partitioned into four semantically distinct intelligence blocks.

Table 27: Feature Taxonomy

Subsystem	Feature Count	Purpose
Agent and Geographic Intelligence	8	Identity, location, commodity context
Financial and Credit Intelligence	10	Informal credit and trust modeling
Supply-Chain Intelligence	10	Logistics and environmental dynamics
Negotiation Intelligence	12	Dialogue and bargaining telemetry
Total	40	Unified economic state representation

This hierarchical organization enables modular experimentation in reinforcement learning, graph learning, language modeling, and economic forecasting tasks.

A.3 Agent and Geographic Intelligence

The first feature block captures participant identity, regional context, and commodity information.

Table 28: Agent and Geographic Variables

Variable	Description
transaction_id	Unique row-level identifier
trajectory_id	Negotiation episode identifier
turn_number	Sequential bargaining turn
distributor_id	Distributor agent identifier
kirana_id	Retailer agent identifier
geographic_tier	Urbanization category
state_context	Regional state context
commodity_type	Negotiated product category

Geographic contexts span twelve Indian states, enabling investigation of regional behavioral heterogeneity and localized economic patterns.

A.4 Financial and Credit Intelligence

The financial subsystem models the informal credit ecosystem commonly referred to as *Udhar*.

Table 29: Financial Intelligence Variables

Variable	Description
historical_trust_score	Longitudinal trust estimate
credit_limit_allocated	Distributor credit allocation
outstanding_udhar	Active debt exposure
repayment_velocity	Historical repayment speed
liquidity_stress_index	Cash-flow pressure indicator
margin_floor	Distributor profitability threshold
payment_modality	Preferred payment mechanism
order_volume_frequency	Historical order intensity
default_risk_probability	Estimated credit default risk
interest_rate_informal	Informal credit interest rate

These variables collectively encode the latent economic state governing bargaining power, credit eligibility, and settlement probability.

A.5 Environmental and Supply-Chain Intelligence

The environmental subsystem models exogenous variables affecting supply-chain operations.

Table 30: Environmental and Logistics Variables

Variable	Description
monsoon_disruption_index	Weather-related disruption
festival_proximity	Demand seasonality indicator
perishability_decay_rate	Inventory degradation rate
fuel_price_volatility	Transportation cost fluctuation
mandi_arrival_shock	Supply arrival deviation
local_inflation_index	Regional inflation signal
warehouse_temperature	Storage condition indicator
competitor_density	Local competition intensity
transportation_mode	Freight mechanism
bottleneck_type	Supply-chain disruption class

These variables create a realistic operational environment in which negotiations occur under uncertainty.

A.6 Negotiation Intelligence and Dialogue Telemetry

The final subsystem captures bargaining dynamics, pricing trajectories, and conversational information.

Table 31: Negotiation Intelligence Variables

Variable	Description
initial_ask_price	Distributor opening price
initial_counter_offer	Retailer opening bid
current_offer_price	Active negotiation price
price_elasticity	Buyer sensitivity metric
negotiation_posture	Behavioral state label
settlement_status	Outcome state
final_settled_price	Final agreement price
transaction_value	Total contract value
json_telemetry	Structured negotiation log
dialogue_raw	Code-switched negotiation utterance
sentiment_score	Conversational sentiment intensity
gp_noise_artifact	Gaussian process perturbation

This telemetry layer enables research at the intersection of strategic decision making, dialogue systems, and behavioral economics.

A.7 Outcome Distribution Statistics

Negotiation outcomes exhibit realistic heterogeneity.

Table 32: Outcome Distribution

Outcome	Proportion
Settled	61.23%
Walked Away	20.47%
In Progress	15.54%
Credit Denied	2.76%

The dominance of successful settlements reflects the repeated-game nature of distributor–retailer interactions, where long-term relationships incentivize cooperation.

A.8 Correlation Structure Verification

To preserve economically meaningful dependencies, KiranaChain incorporates structured copula-based dependency modeling.

Table 33: Representative Correlation Statistics

Variable Pair	Pearson r	Interpretation
Trust Score vs Default Risk	−0.8429	Strong inverse relationship
Liquidity Stress vs Default Risk	0.85	Positive risk dependency
Trust Score vs Credit Limit	0.82	Credit allocation effect
Monsoon Index vs Perishability	0.66	Environmental dependency

These relationships demonstrate successful preservation of domain-informed economic structure within the benchmark.

A.9 Trajectory Statistics

Each negotiation episode is represented as a fixed-length trajectory:

$$\tau = (s_1, a_1, r_1, \dots, s_8, a_8, r_8).$$

Key trajectory characteristics include:

- Fixed horizon length of 8 turns,
- Non-stationary negotiation dynamics,
- Absorbing terminal states,
- Structured concession processes,
- Temporally correlated behavioral perturbations,
- Deterministic trajectory-level dataset splitting.

These properties make the benchmark particularly suitable for reinforcement learning and sequential decision-making research.

A.10 Benchmark Splits

To prevent information leakage, partitioning occurs exclusively at the trajectory level.

Table 34: Official Dataset Splits

Split	Rows	Trajectories	Share
Training	800,000	100,000	80%
Validation	100,000	12,500	10%
Test	100,000	12,500	10%

Formally,

$$\mathcal{T}_{\text{train}} \cap \mathcal{T}_{\text{val}} = \emptyset,$$

$$\mathcal{T}_{\text{train}} \cap \mathcal{T}_{\text{test}} = \emptyset,$$

$$\mathcal{T}_{\text{val}} \cap \mathcal{T}_{\text{test}} = \emptyset.$$

This guarantees zero trajectory leakage across benchmark partitions and ensures fair evaluation protocols for all downstream learning tasks.

A.11 Appendix Summary

The statistical evidence presented throughout this appendix demonstrates that KiranaChain combines large-scale trajectory structure, realistic economic dependency modeling, multilingual conversational intelligence, and rigorous validation procedures within a unified benchmark framework. The resulting dataset provides a reproducible experimental platform for reinforcement learning, graph intelligence, multi-agent systems, supply-chain analytics, behavioral economics, and foundation-model research in decentralized retail environments. ““

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B Generation Parameters

This section documents the probabilistic mechanisms, dependency structures, stochastic processes, and simulation parameters used to generate KiranaChain. The objective is to provide complete transparency regarding benchmark construction and enable independent reproduction of the released dataset.

B.1 Financial Dependency Copula Configuration

The financial subsystem is generated using a Gaussian Copula framework designed to preserve economically meaningful dependencies among credit, liquidity, trust, and risk variables.

Let

$$\mathbf{Z} \sim \mathcal{N}(0, \Sigma_{\text{fin}})$$

where

$$\Sigma_{\text{fin}} \in \mathbb{R}^{10 \times 10}$$

denotes the financial correlation matrix.

The latent Gaussian samples are transformed through the standard normal cumulative distribution function

$$U_i = \Phi(Z_i),$$

and subsequently mapped to target marginal distributions using inverse cumulative density transforms.

Table 35: Representative Financial Correlation Structure

Variable Pair	Target Correlation	Economic Interpretation
Trust Score $\leftrightarrow$ Credit Limit	+0.82	Credit allocation dependency
Trust Score $\leftrightarrow$ Default Risk	-0.78	Creditworthiness relationship
Liquidity Stress $\leftrightarrow$ Default Risk	+0.85	Cash-flow vulnerability
Outstanding Udhhar $\leftrightarrow$ Repayment Velocity	+0.58	Debt accumulation dynamics
Default Risk $\leftrightarrow$ Interest Rate	+0.72	Informal risk premium effect

The correlation matrix is projected onto the nearest positive semi-definite manifold prior to Cholesky decomposition to ensure numerical stability during sampling.

B.2 Environmental Copula Configuration

Environmental variables are generated through an independent Gaussian Copula model representing weather conditions, logistics disruptions, inflationary pressure, and commodity supply shocks.

Table 36: Representative Environmental Correlation Structure

Variable Pair	Target Correlation	Interpretation
Monsoon Index $\leftrightarrow$ Perishability	+0.66	Humidity-driven spoilage
Temperature $\leftrightarrow$ Perishability	+0.71	Physical degradation effect
Fuel Volatility $\leftrightarrow$ Bottlenecks	+0.61	Transportation disruption linkage
Mandi Shock $\leftrightarrow$ Inflation	+0.57	Supply-price transmission
Festival Proximity $\leftrightarrow$ Mandi Shock	-0.38	Pre-festival stocking effect

This structure induces realistic interactions among environmental variables while preserving heterogeneous marginal distributions.

B.3 Marginal Distribution Specifications

Table 37: Marginal Distribution Families

Variable	Distribution	Parameters
Trust Score	Beta	(2.5, 1.8)
Liquidity Stress	Beta	(1.8, 2.5)
Monsoon Index	Beta	(1.5, 3.0)
Perishability Rate	Beta	(2.0, 2.5)
Fuel Volatility	Truncated Normal	(0, 3.5)
Festival Proximity	Uniform Integer	[1, 365]
Temperature	Uniform	[18, 42]
Competitor Density	Poisson-Adjusted Uniform	[0, 20]

B.4 Non-Stationary Markov Transition Model

Negotiation outcomes evolve according to a turn-conditioned Markov process.

The state space is defined as

$$\mathcal{S} = \{\text{In-Progress, Settled, Walked-Away, Credit-Denied}\}.$$

Terminal states are absorbing.

Settlement probability is computed as

$$P_{\text{settled}} = \min(0.08 + 0.09t + 0.12\tau, 0.72).$$

where t denotes negotiation turn and τ denotes trust score.

Walk-away probability follows

$$P_{\text{walk}} = \min\left(0.03 + 0.06\frac{t}{8} + 0.15\sigma, 0.35\right).$$

where σ represents liquidity stress.

Credit-denial probability is given by

$$P_{\text{deny}} = \min(0.01 + 0.08 \max(0, \sigma - 0.5), 0.20).$$

The remaining probability mass is assigned to the continuation state.

B.5 Markov Transition Characteristics

Table 38: Behavioral Properties of the Negotiation Engine

Property	Description
Non-Stationary	Transition probabilities vary with turn index
Trust-Aware	Settlement likelihood increases with trust
Stress-Aware	Walk-away and denial probabilities increase with stress
Absorbing States	Terminal outcomes cannot transition further
Forced Resolution	Remaining trajectories terminate at Turn 8

B.6 Gaussian Process Specifications

To model bounded irrationality and unobserved market friction, each trajectory receives a Gaussian Process perturbation.

The covariance kernel is

$$k(t_i, t_j) = \exp\left(-\frac{|t_i - t_j|^2}{2l^2}\right).$$

with length-scale parameter

$$l = 5.$$

Noise trajectories are sampled according to

$$\mathbf{g} \sim \mathcal{GP}(0, K).$$

The resulting realization is scaled by

$$0.12$$

before injection into negotiation price trajectories.

Table 39: Gaussian Process Hyperparameters

Parameter	Value
Kernel Type	Squared Exponential
Length Scale	5.0
Mean Function	Zero
Scaling Factor	0.12
Trajectory Length	8 Turns
Output Variable	statistical_noise_artifact

The Gaussian Process introduces temporally correlated perturbations that reproduce realistic negotiation irregularities while preserving trajectory coherence.

C Data Dictionary

This section provides the authoritative schema specification for all forty benchmark variables.

C.1 Schema Organization

The KiranaChain state representation is organized into four hierarchical intelligence subsystems.

Table 40: Complete Schema Organization

Subsystem	Columns	Count
Agent and Geographic Intelligence	1–8	8
Financial and Credit Intelligence	9–18	10
Environmental and Supply-Chain Intelligence	19–28	10
Negotiation and Dialogue Intelligence	29–40	12
Total	1–40	40

C.2 Complete 40-Feature Schema

Table 41: Authoritative KiranaChain Feature Dictionary

#	Feature	Type	Description
1	transaction_id	UUID	Unique row identifier
2	trajectory_id	UUID	Negotiation episode identifier
3	turn_number	Integer	Turn index within trajectory
4	distributor_id	String	Distributor agent identifier
5	kirana_id	String	Retailer agent identifier
6	geographic_tier	Categorical	Regional development tier
7	state_context	Categorical	State-level context variable
8	commodity_type	Categorical	Commodity category
9	kirana_historical_trust_score	Float	Historical trust metric
10	distributor_credit_limit_allocated	Float	Credit allocation limit
11	current_outstanding_udhar	Float	Outstanding informal debt
12	repayment_cycle_velocity	Float	Repayment duration estimate
13	kirana_liquidity_stress_index	Float	Liquidity pressure indicator
14	distributor_margin_floor	Float	Minimum margin requirement
15	payment_modality_preference	Category	Payment preference
16	historical_order_volume_frequency	Integer	Historical order frequency
17	default_risk_probability	Float	Credit default estimate
18	interest_rate_informal	Float	Informal lending rate
19	monsoon_disruption_index	Float	Weather disruption metric
20	festival_proximity_coefficient	Integer	Festival distance indicator
21	perishability_decay_rate	Float	Inventory decay coefficient
22	fuel_price_volatility_delta	Float	Fuel price deviation
23	mandi_arrival_volume_shocks	Float	Supply shock metric
24	local_inflation_index_fmcs	Float	Regional inflation indicator
25	ambient_warehouse_temperature	Float	Storage temperature
26	competitor_density_radius	Integer	Local competition intensity
27	transportation_mode	Category	Freight modality
28	supply_chain_bottleneck_type	Category	Logistics disruption category
29	distributor_initial_asking_price_per_unit	Float	Opening ask price
30	kirana_initial_counter_offer_per_unit	Float	Opening bid price
31	current_turn_offer_price	Float	Active negotiated price
32	price_elasticity_demanded	Float	Buyer sensitivity coefficient
33	negotiation_posture	Category	Behavioral negotiation state
34	settlement_status	Category	Resolution outcome state
35	final_settled_price_per_unit	Float	Final agreement price
36	total_transaction_value_inr	Float	Gross transaction value
37	llguidance_constrained_json_log	JSON	Structured telemetry log
38	code_switched_dialogue_raw	Text	Negotiation utterance
39	sentiment_intensity_score	Float	Sentiment polarity measure
40	statistical_noise_artifact	Float	Gaussian process perturbation

C.3 Descriptive Statistical Summary

Table 42: Selected Benchmark Statistics

Metric	Mean	Std. Dev.
Trust Score	0.5812	0.2146
Credit Limit (INR)	139,520	53,703
Outstanding Udhar (INR)	48,112	32,152
Liquidity Stress	0.4183	0.2215
Default Risk	0.4014	0.1603
Initial Ask Price (INR)	85.50	41.27
Initial Counter Offer (INR)	64.72	34.11
Current Offer Price (INR)	75.58	37.84
Sentiment Intensity	-0.0758	0.2821

C.4 Appendix Closing Remarks

This appendix provides the methodological foundation of KiranaChain, including the statistical generation process, dependency structures, probabilistic assumptions, trajectory construction methods, validation procedures, and schema specifications. Together, these details ensure transparency, reproducibility, and independent verification of the benchmark.

A key strength of KiranaChain is its ability to support multiple research areas within a unified environment, including reinforcement learning, multi-agent systems, graph intelligence, negotiation modeling, supply-chain analytics, credit-risk assessment, and multilingual conversational AI. The trajectory-centric design enables researchers to study interactions across economic, relational, and linguistic dimensions simultaneously.

Methodologically, KiranaChain combines Gaussian copula dependency modeling, non-stationary Markov dynamics, and Gaussian Process perturbations to generate economically grounded synthetic environments. This approach balances realism, scalability, and reproducibility while preserving meaningful relationships among agents, transactions, environmental factors, and negotiation outcomes.

The benchmark is intended as a foundation for future research and expansion. Potential extensions include larger multi-agent market ecosystems, long-horizon strategic interactions, SKU-level retail intelligence, multilingual and multimodal communication settings, and hybrid benchmarks that incorporate real-world commercial data.

Overall, KiranaChain contributes toward economically grounded AI benchmarking by providing one million trajectory-based observations that capture decentralized economic interactions. We hope it serves as a valuable resource for advancing research in reinforcement learning, graph intelligence, agentic systems, economic modeling, and multilingual reasoning.